Original BK count criteria, directly over finite fields

Result. Primes r=2 and r=3, every nonsingular general Weierstrass model over every finite field F, every finite extension field K with its actual F-algebra structure and cardinality q^k. Retain the source hypotheses r dividing the actual rational point-group cardinality and r not dividing q−1. Exactly r² actual K-points killed by r, including the identity, iff r divides q^k−1. No supplied auxiliary algebraic closure, embedding, scalar tower, coprimality, rank, determinant, pairing or normal-form premise. The qualified source survives these checks. [pcc:unverified]

Prime two	q−1 condition → characteristic two actual count 1 or 2 ≠ required 4

Prime three	canonical closure of chosen K original source theorem → count 9 iff 3 | q^k−1

Prime three. Construct the canonical algebraic closure of the chosen K inside the proof. Mathlib already supplies the canonical F-algebra and scalar tower. These discharge the auxiliary parameters of the preceding original-source prime-three criterion. That proof handles arbitrary nonsingular models and characteristic three, where the actual torsion count is one or three and both nine-point count and cyclotomic divisibility are false. The public statement quantifies only F and K. [pcc:unverified]

Prime two. The source q−1 condition forces 2 dividing q. Finite additive Cauchy and no zero divisors force characteristic two. Existing actual nonzero killed-two point uniqueness now gives a one-or-two count over every characteristic-two field, with no algebraic closure premise. The actual subgroup is finite and never has four points. Cyclotomic divisibility is false for positive k, while the chosen field cardinality excludes k=0. The public source point-count divisibility premise is preserved, although the prime-two proof does not need it. [pcc:unverified]

Actual controls and prior work. Credited characteristic-two models have discriminant1 and actual killed-two subgroup counts2 and1 over every characteristic-two field. An actual point of order two in the ordinary model makes its rational group cardinality even over every finite such field. Previous two-torsion geometry, general three-torsion rank, actual normalization, characteristic-three classification and original-source criterion are reused, not presented as new. [pcc:unverified]

Primary source. BK1998 Theorem1/proof/Remarks pp142–143, DOI10.1007/s001459900040. Original count includes the identity; the inclusion paraphrase remains coprime-qualified. Selected source sections and pinned Mathlib closure/tower APIs read; no full-source-read, novelty, source refutation or corpus promotion. KB/Conductor unavailable, bounded primary/repository/library fallback recorded. [pcc:unverified]

Validation and remaining scope. Whole gate7/0failures/0trusted native exceptions; complete statement binding/recursive source-config pins. All31attempt-provenance files retained losslessly in16objects; five gate records. Failed manual-composition attempt retained; successful proof reuses canonical instances. Pinned Lean4.35.0/Mathlib56a00585…, standard foundations only. Exact local PCC replay is separate from signed standing. Other primes≥5, canonical Weil, full all-k p<500 census, formal supersingularity/trace classification and original fourteen-item gaps remain open. No native experiment, paid compute, upload, holes/custom axioms/native evaluator/raised limits/new global flags/waivers. [pcc:unverified]